Supplementary Material

Glossary and intermediate results:

1. CSV File: A Comma-Separated Values file format used to store tabular data.
2. readtable Function: MATLAB function that reads data from a text file into a table.
3. grp2idx Function: Converts categorical variables into numeric indices, essential for machine learning models that require numerical inputs.
4. cvpartition Function: Used to create a cross-validation partition to split the dataset into training and testing sets.
5. SMOTE (Synthetic Minority Over-sampling Technique): To balance class distribution in a dataset by artificially creating new instances from the minority class.
6. fitcnb Function: Fits a Naive Bayes classifier to the training data in MATLAB.
7. confusionmat Function: Generates a confusion matrix which is a summary of prediction results on a classification problem.
8. perfcurve Function: Used to compute the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC).
9. Intermediate Results: The processing steps reveal intermediate outcomes such as the transformation of categorical data into numerical indices, the division of the dataset into training and testing sets, and the generation of synthetic samples for balancing the dataset.

Implementation Details:

1. Handling Data Imbalance: The decision to use SMOTE is based on the understanding that imbalanced classes can lead to biased models. SMOTE helps in creating a more balanced dataset, enhancing the model's ability to generalize.
2. Performance Metrics Choice: Metrics like accuracy, precision, recall, and F1-score provide a comprehensive view of the model's performance. The ROC-AUC metric is particularly useful in evaluating the trade-offs between true positive rate and false positive rate.
3. Dataset Splitting Strategy: Using a holdout method (20% of data for testing) provides a balance between having enough data to train the model and enough to test its performance.
4. Table-to-Array Conversion: Since some MATLAB functions require data in an array format, converting tables to arrays ensures compatibility and smooth functioning of the script.